



E-COMMERCE SALES ANALYSIS FOR DATA-DRIVEN DECISION MAKING

- **Presented By :-** Mundru Vedavedyaram
- **Project Type :-** Data Analytics & Business Intelligence

PROJECT OVERVIEW

Introduction

- The purpose of this project was to analyse sales data and generate actionable business insights using the Sample Superstore dataset.

Objectives

- Analyse sales performance over time
- Identify best-selling products and categories
- Understand customer purchasing behaviour
- Forecast future sales trends
- Identify business growth opportunities

DATASET OVERVIEW

Dataset Information

Key Features

- Order Date
- Sales
- Profit
- Quantity
- Customer Information
- Product Information
- Geographic Information ‘

Business Areas Covered

- Sales Analysis
- Customer Analysis
- Product Performance
- Geographic Performance
- Forecasting

Dataset Name

- Sample Superstore Dataset

Dataset Size

- 9,994 Records
- 21 Columns

DATA CLEANING AND PREPROCESSING

Activities Performed

- Missing Value Detection
- Duplicate Record Detection
- Date Format Conversion
- Data Type Validation
- Outlier Identification

Findings

- No missing values found
- No duplicate records identified
- Date columns converted successfully
- Dataset ready for analysis

MONTHLY & YEARLY SALES TRENDS

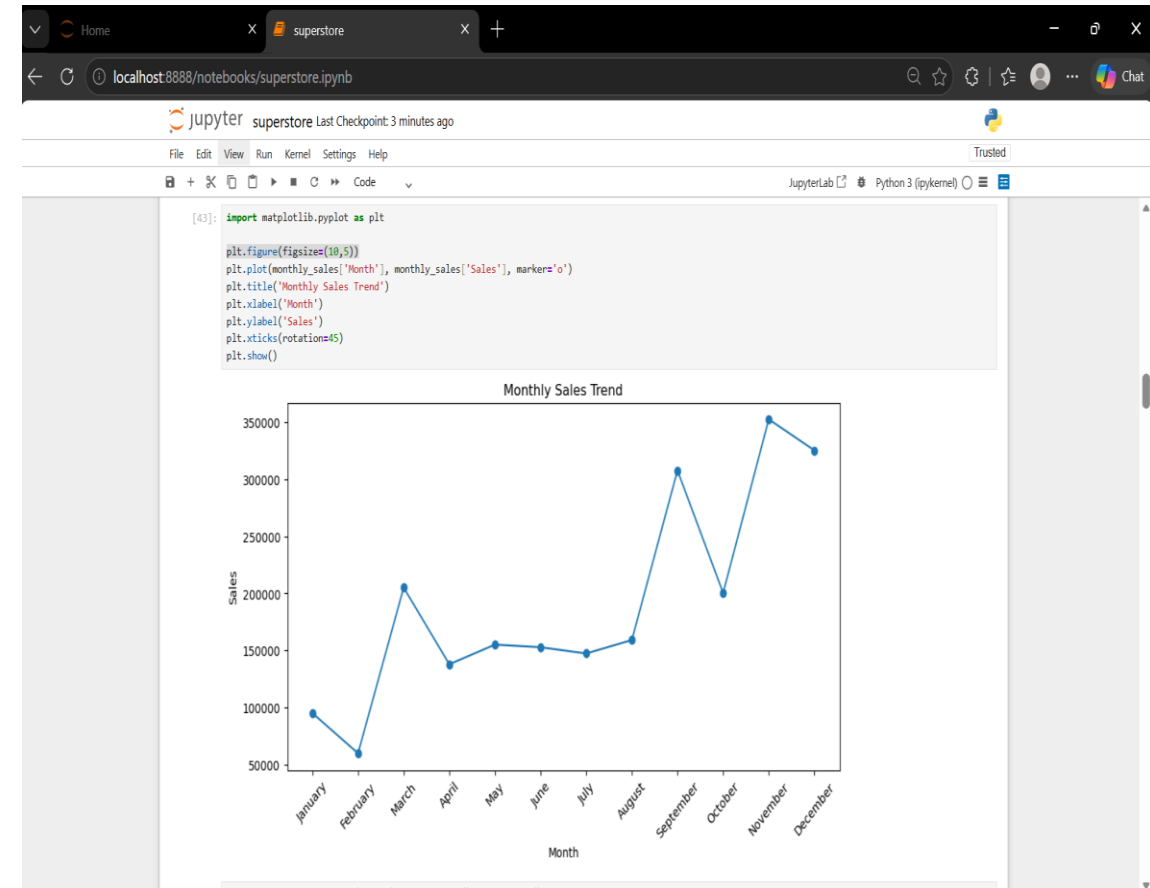
Sales Trend Analysis

Key Findings

- Sales fluctuate throughout the year
- Seasonal patterns are visible
- Some months generate significantly higher revenue
- Overall sales performance shows positive growth

Business Impact

- Understanding seasonal trends helps optimise:
- Inventory planning
- Marketing campaigns
- Resource allocation



BEST-SELLING PRODUCTS & CATEGORIES

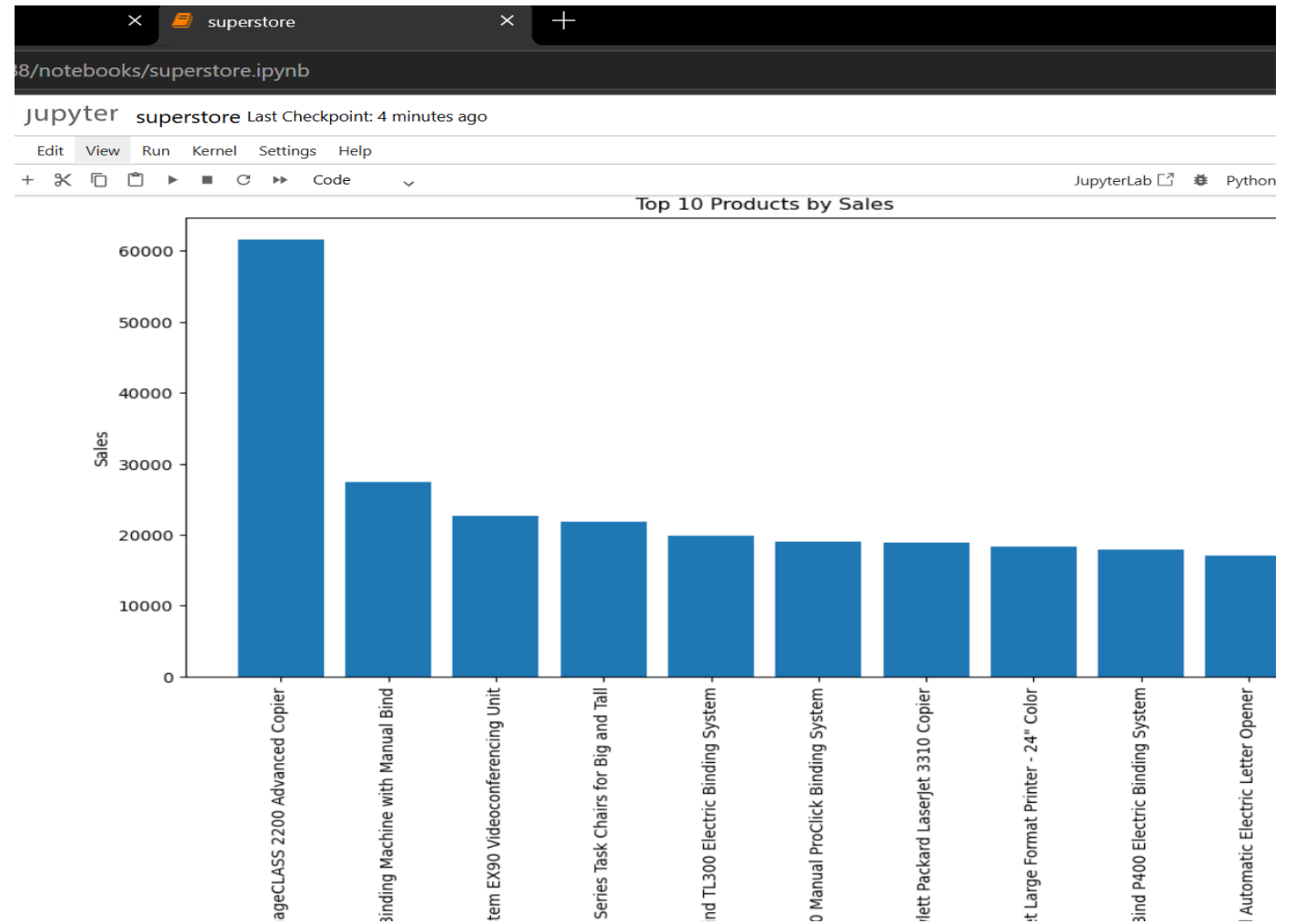
Product Performance Analysis

Findings

- The technology category generated the highest revenue
- Office Supplies contributed significantly
- Furniture maintained stable performance

Benefits

- Identify high-demand products
- Improve inventory planning
- Focus marketing on top-performing products



GEOGRAPHIC SALES PERFORMANCE

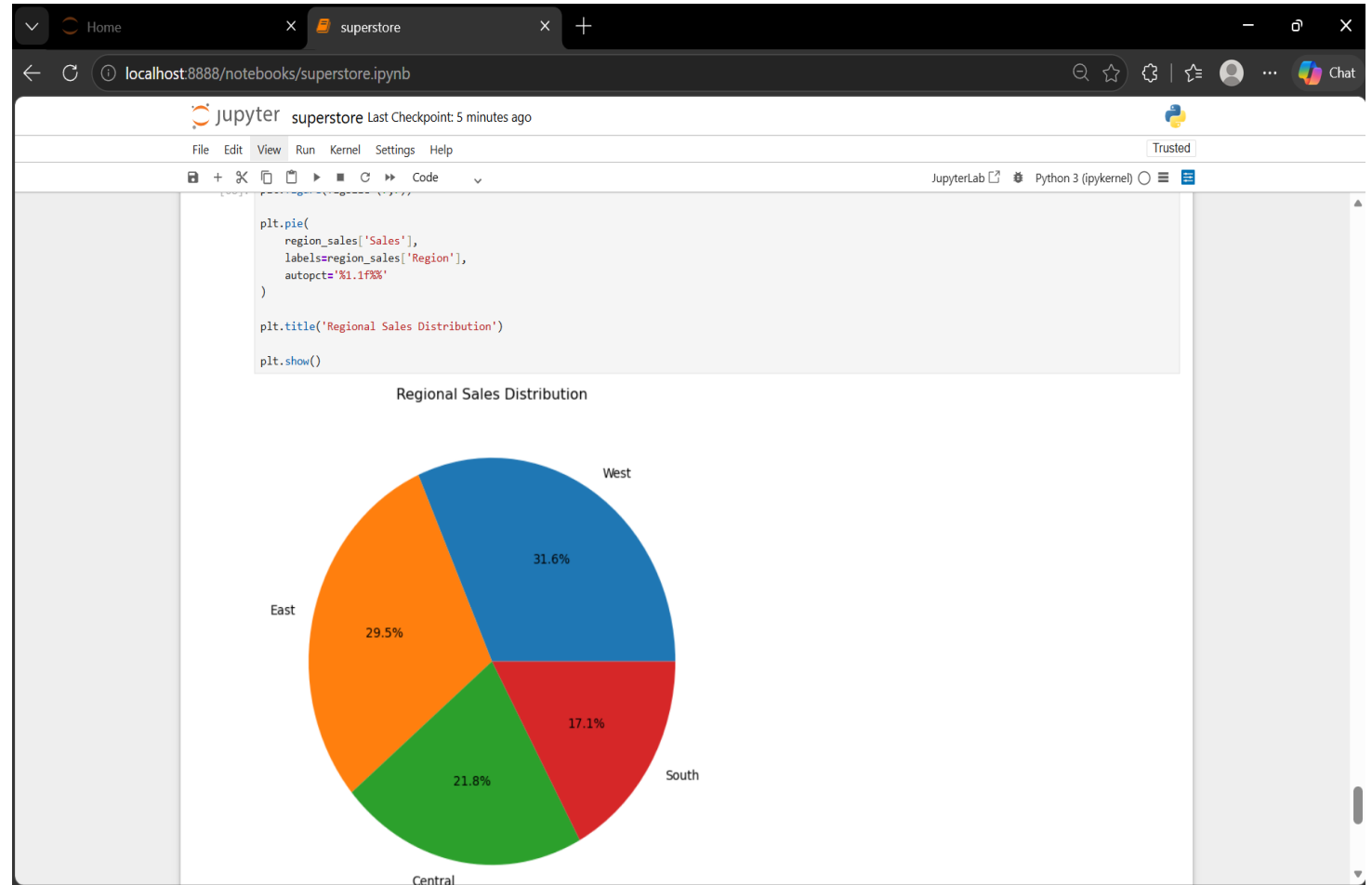
Regional Analysis

Findings

- Certain states generated significantly higher sales
- Urban markets showed stronger demand
- Regional sales performance varied considerably

Business Opportunities

- Expand operations in high-performing regions
- Improve marketing in underperforming areas
- Optimise logistics and distribution



CUSTOMER SEGMENTATION (RFM ANALYSIS)

Customer Behaviour Analysis

At-Risk Customers

Segments Identified

- Customers requiring re-engagement

High-Value Customers

- Recent, frequent, and high-spending customers

Lost Customers

- Inactive customers with low engagement

Loyal Customers

Benefits

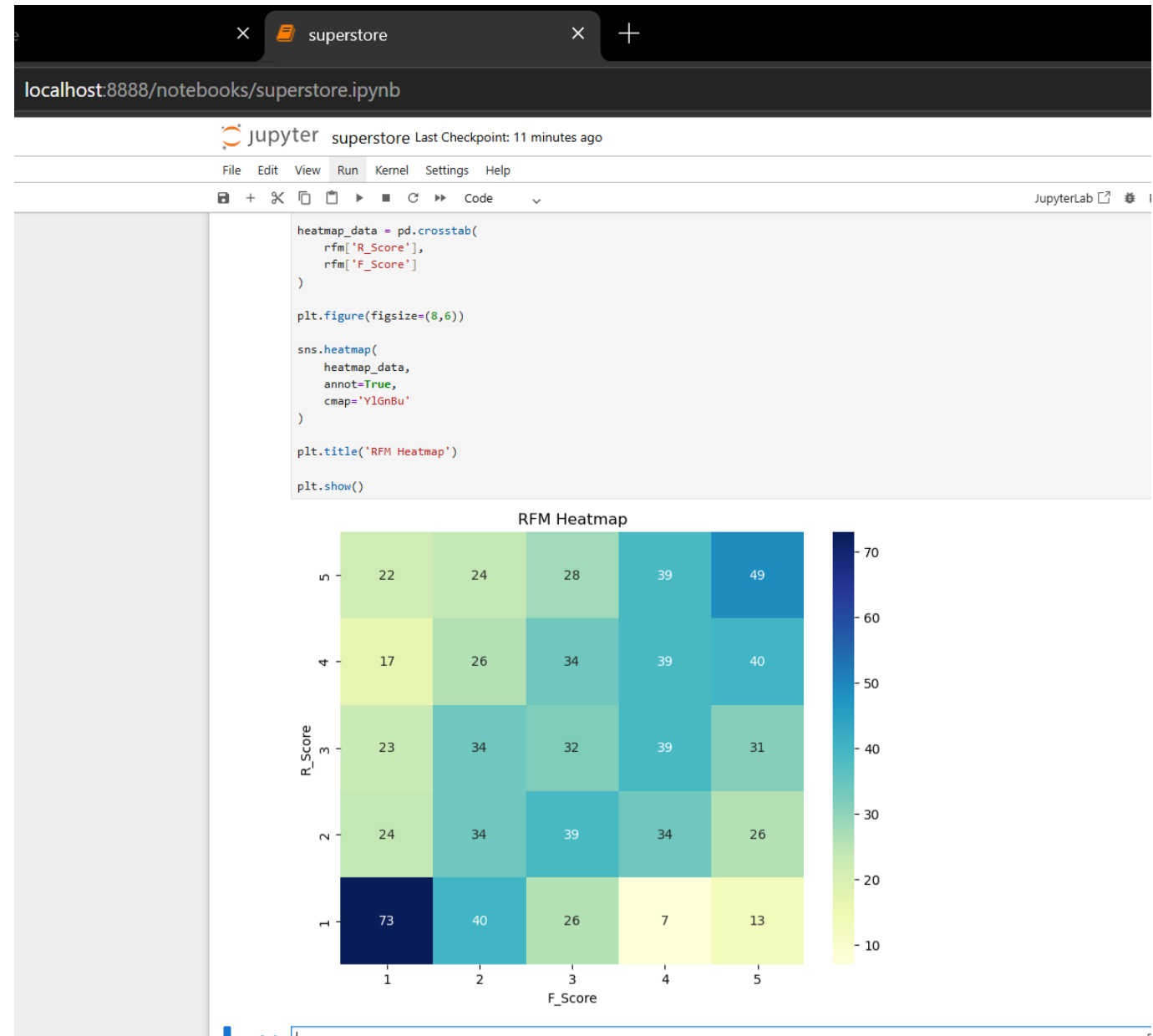
- Regular repeat buyers
- Personalised marketing

Potential Loyalists

- Improved retention

- Customers with strong future potential

- Increased customer lifetime value



SALES FORECASTING

Future Sales Prediction

Forecasting Model Used

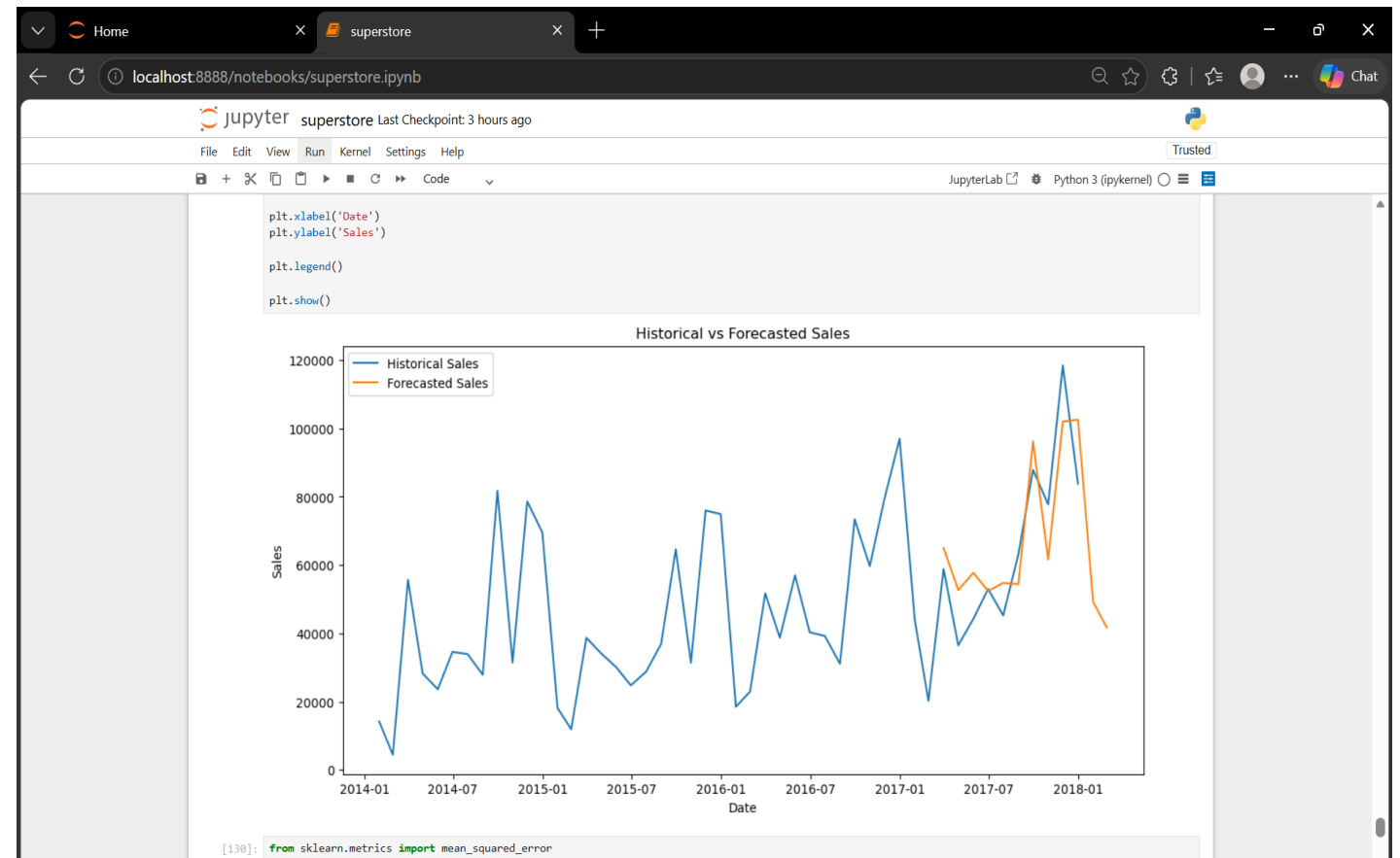
- Holt-Winters Exponential Smoothing

Why This Model?

- Captures trends
- Handles seasonality
- Suitable for monthly sales forecasting

Forecast Insights

- Sales expected to follow historical trends
- Seasonal fluctuations continue
- Useful for demand planning



BUSINESS OPPORTUNITIES

Key Opportunities Identified

1. Technology Product Expansion

- Increase inventory
- Launch premium products

2. Regional Growth Strategy

- Focus on high-performing markets
- Improve low-performing regions

3. Customer Retention Programs

- Loyalty rewards
- Personalized promotions

4. Cross-Selling & Upselling

- Product bundles
- Complementary recommendations

RECOMMENDATIONS

Strategic Recommendations

Inventory Management

- Increase stock for high-demand products
- Prepare for forecasted peak periods

Marketing Strategy

- Personalized campaigns using customer segmentation
- Region-specific promotions

Customer Retention

- Loyalty programs
- Re-engagement campaigns

Forecast-Driven Planning

- Use forecast results for future business decisions

CONCLUSION

Key Achievements

- Analysed sales trends and product performance
- Identified valuable customer segments
- Forecasted future sales demand
- Discovered business growth opportunities

Final Takeaway

- Data-driven decision-making can significantly improve:
 - Revenue Growth
 - Customer Satisfaction
 - Inventory Optimisation
 - Business Performance

Thank You

Questions & Answers

Presented By :-

Mundru Vedavedyaram